IV.5 Newton's Method for Finding Extrema (1 dimension)

Our goal is to find a <u>local max/min</u> of a function $f(x):\mathbb{R}\to\mathbb{R}$, where $f''(x)$ exists.

Recall from Analytic Calculus 1:

1. If $f(x)$ has a local max/min at a point x_0, then $f'(x_0)=0$.
   (We call such an x_0 a *critical point* or *critical number* of $f(x)$)
2. (Second derivative test) Let x_0 be a critical number of $f(x)$.
   - If $f''(x_0)<0$ then $f(x)$ has a local maximum at x_0.
   - If $f''(x_0)>0$ then $f(x)$ has a local minimum at x_0.
   - If $f''(x_0)=0$ then this test is inconclusive.

By 1., we must find solutions to $f'(x) = 0$. We can find such solutions using the bisection method, Netwon's method or the secant method, applied to $f'(x)$.

a) Newton's Method.

Recall the formula in Newton's method to find solutions to $g(x) = 0$:

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)} \qquad (1)$$

Because we want to solve $\boxed{f'(x) = 0,}$ we choose $\boxed{g(x) = f'(x).}$ Then $g'(x) = f''(x)$, and (1) becomes *Newton's formula for finding critical points* iteratively:

$$\boxed{x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}} \qquad (2)$$

Example 1: Use Newton's method to find the minimizer of

$$f(x) = \tfrac{1}{2}x^2 - \sin x.$$

$$\boxed{x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}}$$

Stop when $|x_{k+1} - x_k| < 10^{-4}$.

Sol: 1. Find a critical point numerically.

$$f'(x) = x - \cos x \quad \text{and} \quad f''(x) = 1 + \sin x$$

So rule (2) becomes

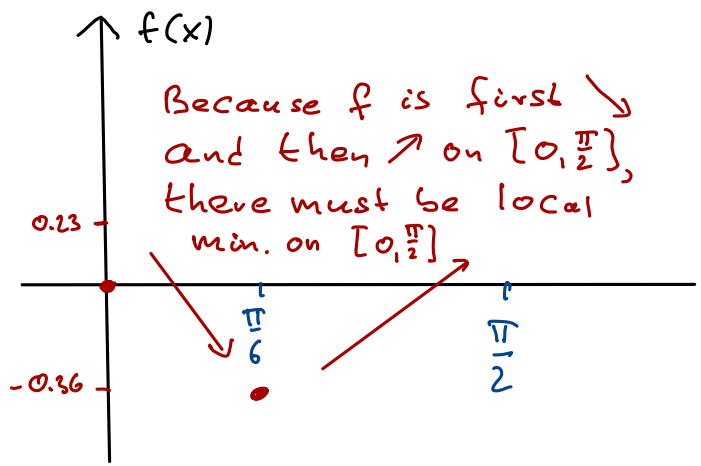
$$x_{k+1} = x_k - \frac{x_k - \cos x_k}{1 + \sin x_k}. \qquad (3)$$

What should our starting point x_0 be ? Let's use "trial and error"

$x = 0$: $\quad f(0) = 0 - \sin(0) = \underline{\underline{0}}$

$x = \frac{\pi}{6}$: $\quad f(\frac{\pi}{6}) = \frac{1}{2}\cdot\left(\frac{\pi}{6}\right)^2 - \sin\left(\frac{\pi}{6}\right) = \underline{\underline{-0.36}}$

$x = \frac{\pi}{2}$: $\quad f(\frac{\pi}{2}) = \frac{1}{2}\left(\frac{\pi}{2}\right)^2 - \sin\left(\frac{\pi}{2}\right) = \underline{\underline{0.23}}$



So we expect a local min inside the interval $\left[0, \frac{\pi}{2}\right]$. We choose x_0 from inside this interval, for example, $x_0 = 0.5$ [alternatively, we can choose $x_0 = 1$]

We compute the first iterations using (3): $x_{k+1} = x_k - \dfrac{x_k - \cos(x_k)}{1 + \sin(x_k)}$

$$x_1 = x_0 - \frac{x_0 - \cos x_0}{1 + \sin x_0} = 0.5 - \frac{0.5 - \cos(0.5)}{1 + \sin(0.5)} = \underline{\underline{0.7552}}$$

$$x_2 = x_1 - \frac{x_1 - \cos x_1}{1 + \sin x_1} = 0.7552 - \frac{0.7552 - \cos(0.7552)}{1 + \sin(0.7552)} = \underline{\underline{0.7391}}$$

$$x_3 = x_2 - \frac{x_2 - \cos x_2}{1 + \sin x_2} = 0.7391 - \frac{0.7391 - \cos(0.7391)}{1 + \sin(0.7391)} = \underline{\underline{0.7390}}$$

$$x_4 = x_3 - \frac{x_3 - \cos x_3}{1 + \sin x_3} = 0.7390 - \frac{0.7390 - \cos(0.7390)}{1 + \sin(0.7390)} = \underline{\underline{0.7390}}$$

Because $x_4 = x_3$ up to 4 decimal digits, then $|x_4 - x_3| < 10^{-4}$ and we stop.

So we take $\boxed{x^* = 0.7390}$ as a <u>critical point of</u> $f(x)$.

2. Use the second derivative test: $\Leftrightarrow$ to test for Max/Min

$$f''(x) = 1 + \sin x \qquad (4)$$

Then $f''(x^*) = 1 + \sin(0.7390) = 1.6735 > 0 \Rightarrow$ Min

So $x^* = 0.7390$ is a (local) minimizer of $f(x)$.

The local min. value is $\boxed{f(0.7390) = -0.4005}$

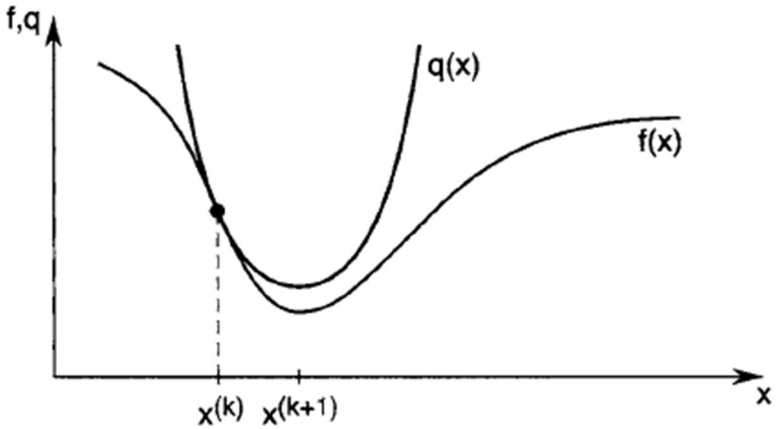
Remark: By (4), $f''(x) = 1 + \sin x \geq 0$ for <u>all</u> x.

This means that the graph of $f(x)$ is <u>convex upwards</u> $(-\infty, \infty)$. In this case, a local min value is also the absolute min value.

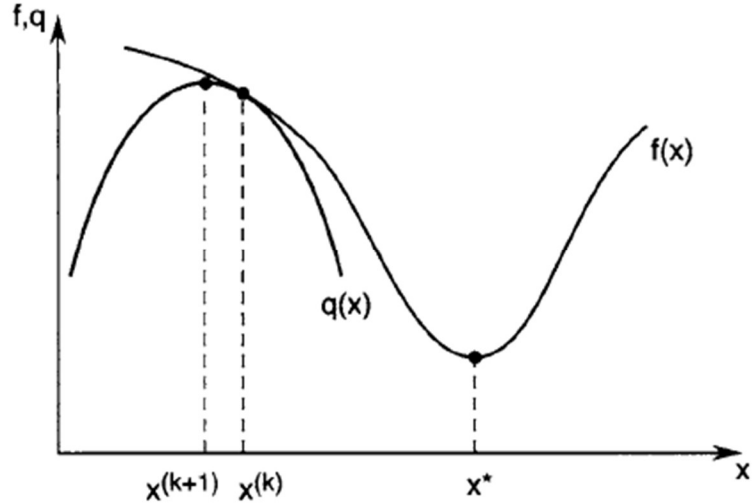
So $x^* = 0.7390$ is the (absolute) minimizer of $f(x)$.

and the absolute min. value is $f(0.7390) = -0.4005$

<u>Remark:</u> In a <u>minimization</u> problem, Newtons method works best when $f''(x) > 0$ (the graph is convex upward) around the iterations x_k.



$f''(x) > 0$ around x_k



$f''(x) < 0$ around x_k

In a <u>maximization</u> problem, Newtons method works best when $f''(x) < 0$ (the graph is concave downward) around the iterations x_k.

b) The secant Method.

Recall the formula in the secant method to find solutions to $g(x) = 0$:

$$x_{k+1} = x_k - \frac{(x_k - x_{k-1})g(x_k)}{g(x_k) - g(x_{k-1})} \qquad (5)$$

Because we want to solve $f'(x) = 0$, we choose $\boxed{g(x) = f'(x)}$, and (5) becomes *the secant formula for finding critical points* iteratively:

$$\boxed{x_{k+1} = x_k - \frac{(x_k - x_{k-1})f'(x_k)}{f'(x_k) - f'(x_{k-1})}} \qquad (6)$$

Example 2: Use the secant method to find the minimizer of

$$f(x) = \tfrac{1}{2}x^2 - \sin x.$$

Stop when $|x_{k+1} - x_k| < 10^{-4}$.

Sol: Apply rule (6), $$x_{k+1} = x_k - \frac{(x_k - x_{k-1})f'(x_k)}{f'(x_k) - f'(x_{k-1})}$$

with $f'(x) = x - \cos x$. We use starting points $\boxed{x_0 = 0.5 \text{ and } x_1 = 1}$

both lie in $[0, \frac{\pi}{2}]$

k	x_k	$f'(x_k)$
0	0.5	-0.3776
1	1.0	0.4597
2	0.7255	-0.02267
3	0.7384	-0.001147
4	0.7390	0.0000249
5	0.7390	

$k=1$: $$x_2 = x_1 - \frac{(x_1 - x_0)f'(x_1)}{f'(x_1) - f'(x_0)}$$

$$= 1 - \frac{(1-0.5)\cdot 0.4597}{0.4597 - [-0.03776]} = 0.7255$$

$k=2$: $$x_3 = x_2 - \frac{(x_2 - x_1)f'(x_2)}{f'(x_2) - f'(x_1)}$$

$$= 0.7255 - \frac{(0.7255 - 1)(-0.02267)}{(-0.02667 - 0.4597)}$$

$$= 0.7384$$

$|x_5 - x_4| < 10^{-4} \Rightarrow$ stop. $x^* = 0.7390$ Critical number

IV.6 Newton's Method for Finding Extrema (many dimensions)

Let $f(\mathbf{x}) = f(x_1, x_2, \ldots, x_n)$ be a function of several variables, where the the second order partial derivatives $\dfrac{\partial^2 f}{\partial x_i \, \partial x_j}$ all exist an are continuous.

We recall that the **gradient** of f at a point $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is the vector consisting of first order partial derivatives.

$$\nabla f(\mathbf{x}) = \left\langle \frac{\partial f}{\partial x_1}(\mathbf{x}), \ \frac{\partial f}{\partial x_2}(\mathbf{x}), \ \ldots\ldots \ , \ \frac{\partial f}{\partial x_n}(\mathbf{x}) \right\rangle$$

(Gradient is the equivalent of the derivative $f'(x)$ of an ordinary function)

The equivalent of the derivative $f''(x)$ of an ordinary function is now an $n \times n$ matrix, called the ***Hessian Matrix,***

$$D^2 f(\vec{x}) = \nabla^2 f(\mathbf{x}) = F(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_3} & \cdots\cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_3} & \cdots\cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_3 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_3 \partial x_2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_3^2} & \cdots\cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_3 \partial x_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_3} & \cdots\cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

Because we assume that the second order partial derivatives are continuous, then

$$\boxed{\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}} \quad \text{for all } i, j$$

so that this matrix is symmetric.

Example 3: Consider $f(\mathbf{x}) = f(x,y) = x^4 + y^4 - 4xy + 2$

Here, $\dfrac{\partial f}{\partial x}(x,y) = 4x^3 - 4y$ and $\dfrac{\partial f}{\partial y}(x,y) = 4y^3 - 4x$

So the gradient is

$$\nabla f(x,y) = \langle 4x^3 - 4y, 4y^3 - 4x \rangle$$

The second order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = 12x^2 \qquad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -4 \qquad \frac{\partial^2 f}{\partial y^2} = 12y^2$$

Therefore the Hessian matrix is

$$\nabla^2 f(x,y) = F(x,y) = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x \partial y} \\ \dfrac{\partial^2 f}{\partial y \partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}$$

Example 4: Consider $f(\mathbf{x}) = f(x_1, x_2, x_3) = x_1 \sin(2x_2 + x_3^2)$

chain rule

1. Find the gradient.

$$\frac{\partial f}{\partial x_1} = \sin(2x_2 + x_3^2) \qquad \frac{\partial f}{\partial x_2} = x_1 \cdot \cos(2x_2 + x_3^2) \cdot 2 \qquad \frac{\partial f}{\partial x_3} = x_1 \cdot \cos(2x_2 + x_3^2) \cdot 2x_3$$

$$\nabla f(x_1, x_2, x_3) = \langle \sin(2x_2 + x_3^2), 2x_1 \cdot \cos(2x_2 + x_3^2), 2x_1 x_3 \cos(2x_2 + x_3^2) \rangle$$

2. The Hessian matrix is:

$$F(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 0 & 2\cos(2x_2 + x_3^2) & 2x_3 \cos(2x_2 + x_3^2) \\ 2\cos(2x_2 + x_3^2) & -4x_1 \sin(2x_2 + x_3^2) & -4x_1 x_3 \sin(2x_2 + x_3^2) \\ 2x_3 \cos(2x_2 + x_3^2) & -4x_1 x_3 \sin(2x_2 + x_3^2) & 2x_1 \cos(2x_2 + x_3^2) - 4x_1 x_3^2 \sin(2x_2 + x_3^2) \end{bmatrix}$$

(product rule)

Recall from the beginning of the course: Consider a function of two variables, $f(\mathbf{x}) = f(x, y)$. We want to find local max/min values.

1. *(Necessary condition)* If $f(x, y)$ has a local extremum at the point (x_0, y_0) then

$$\nabla f(x_0, y_0) = 0$$

$\frac{\partial f}{\partial x}(x_0, y_0) = 0$ <u>and</u>

$\frac{\partial f}{\partial y}(x_0, y_0) = 0$

((x_0, y_0) is called a **critical point** of f.)

A local extremum can only occur at a critical point.

2. *(Sufficient condition)* Check the sign of the discriminant at each <u>critical point</u> (x_0, y_0):

$$D(x_0, y_0) = \underbrace{\begin{vmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{xy}(x_0, y_0) & f_{yy}(x_0, y_0) \end{vmatrix}}_{\text{This is the determinant of the Hessian matrix}}$$

a) Case $D(x_0, y_0) > 0$:

If $f_{xx}(x_0, y_0) > 0$ or $f_{yy}(x_0, y_0) > 0$: local min at (x_0, y_0)

If $f_{xx}(x_0, y_0) < 0$ or $f_{yy}(x_0, y_0) < 0$: local max at (x_0, y_0)

Case $D(x_0, y_0) < 0$: f has **no** local max/min at (x_0, y_0)

((x_0, y_0) is a saddle point)

We now want to generalize this to a function of many variables.